



SPLICE PLAYGROUND

Visualize Splicing Into Structure

DNA editing, transcript consequences, and protein structure prediction in one
continuous flow.

[Start](#)



2026S – Graduation Project2

Splice Playground

Make your own mutant

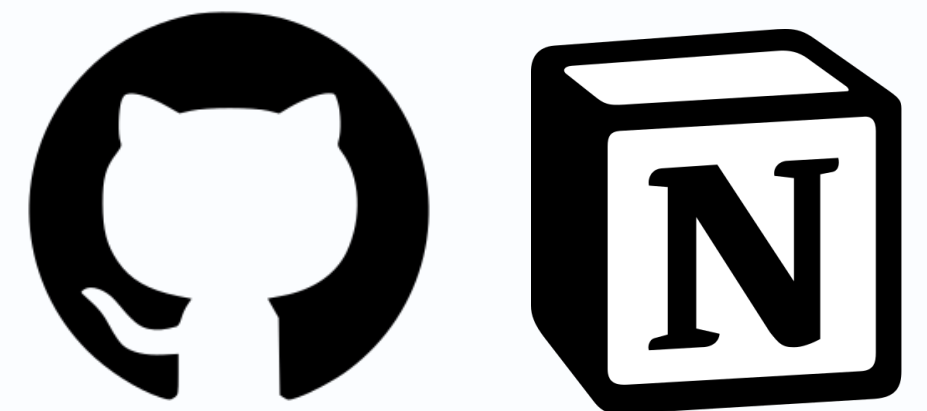
AI01분반

4조: 강민준, 김현우, 남윤정, 이정균, 최진범

Department of AI-Software

Outline

1. Concept Review
2. Splice AI training result
3. Service Architecture
4. DB Construction
5. Progress & Demo
6. Future Work & Mentoring
7. 창업플랜

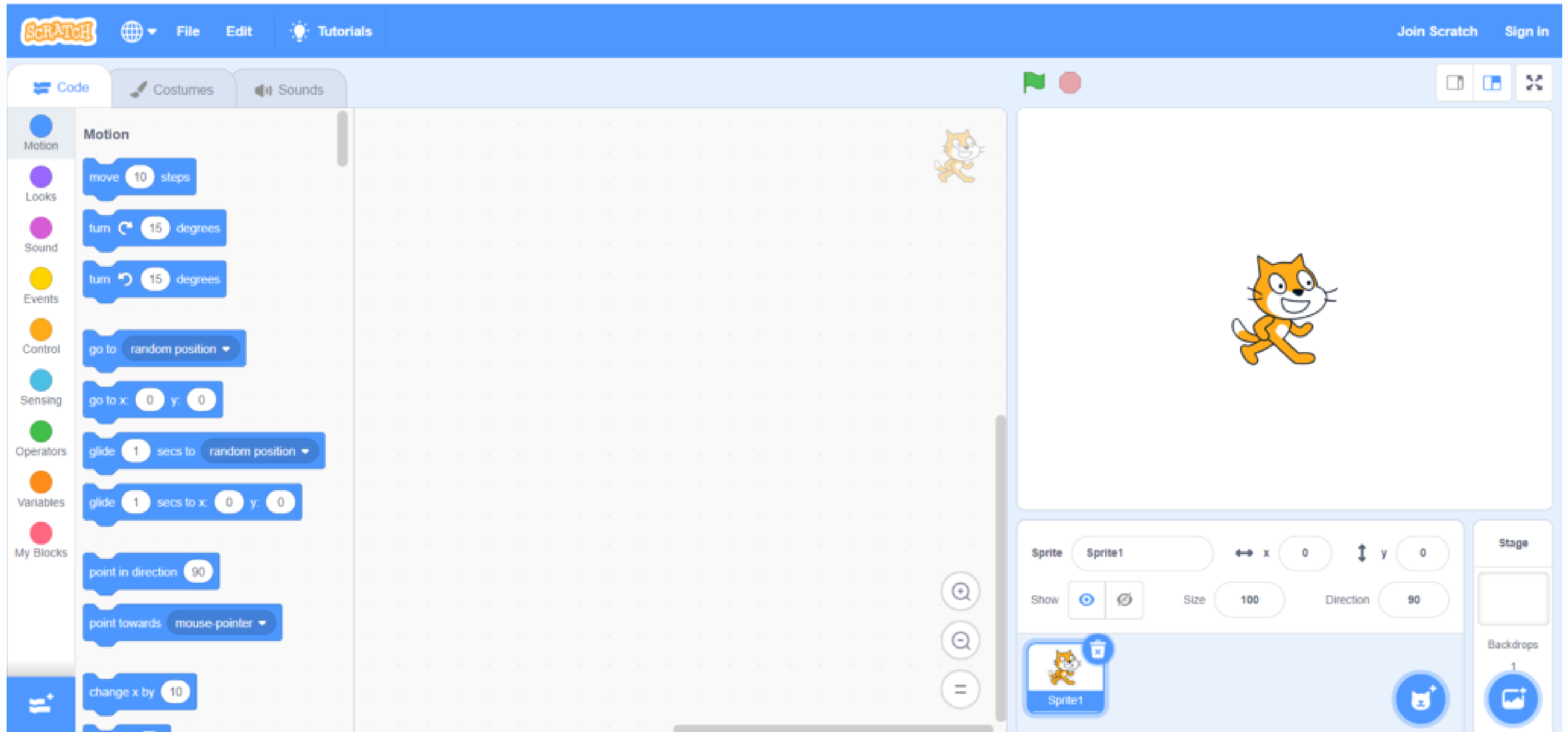


1. Concept Review

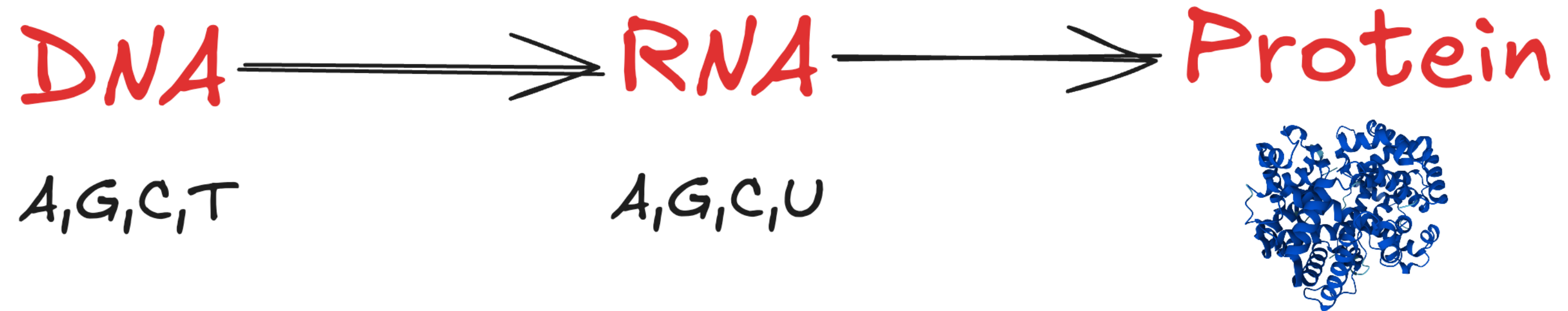
Concept Review



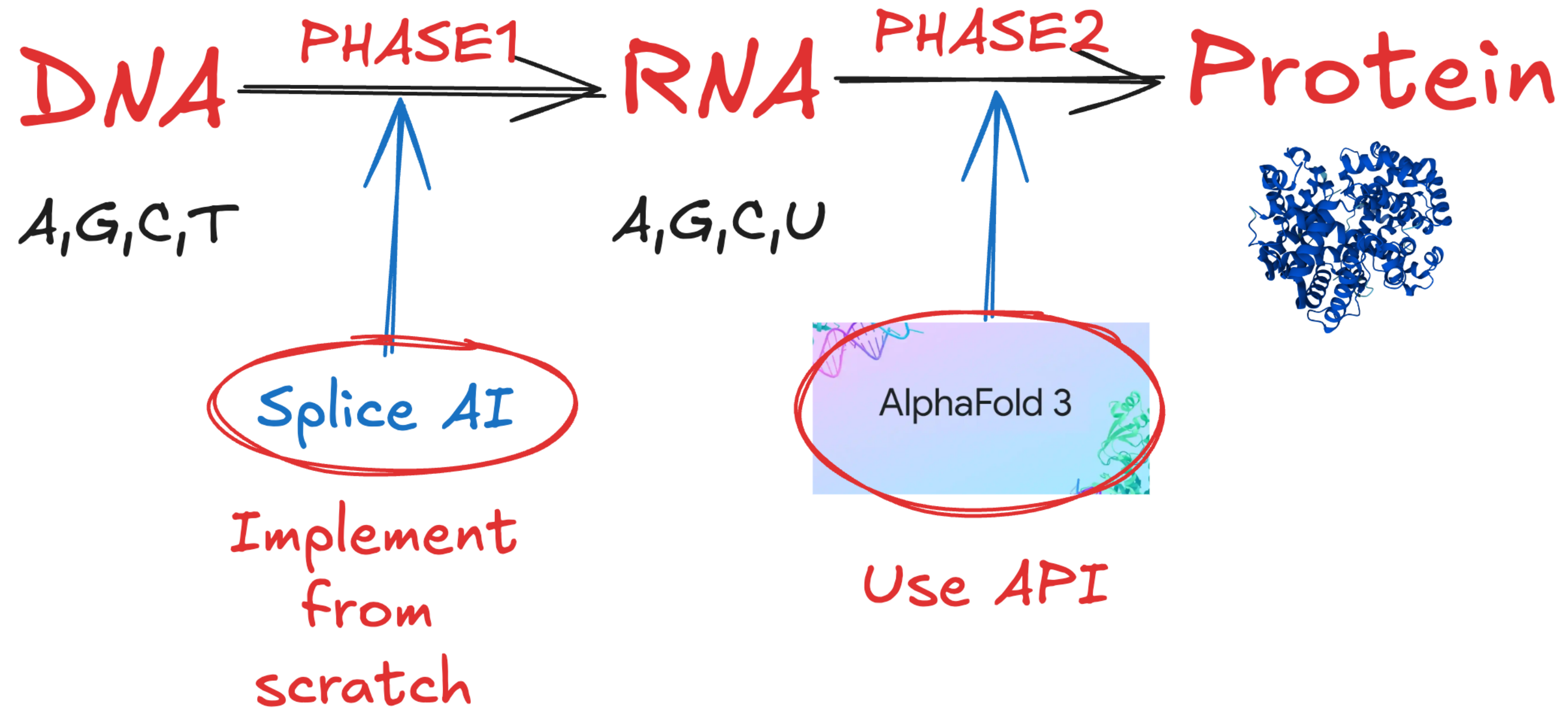
Concept Review



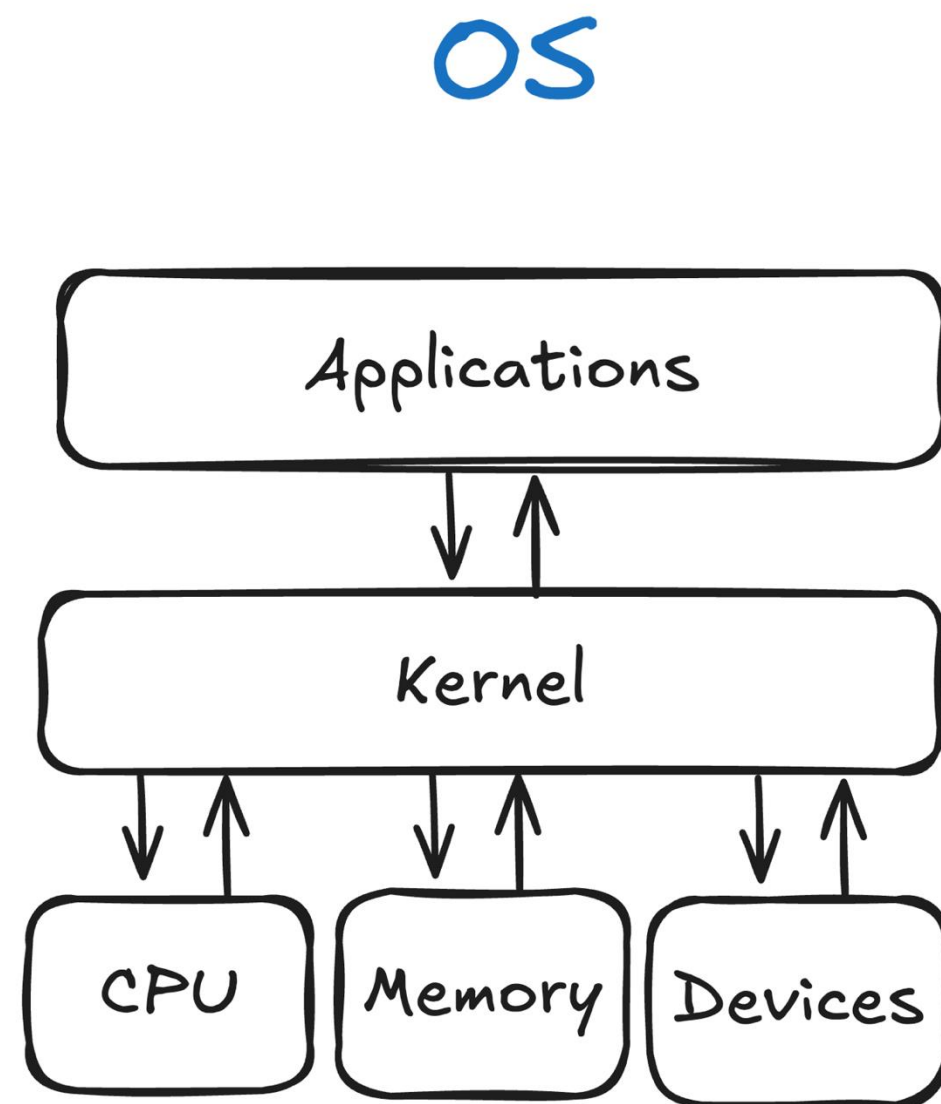
Concept Review



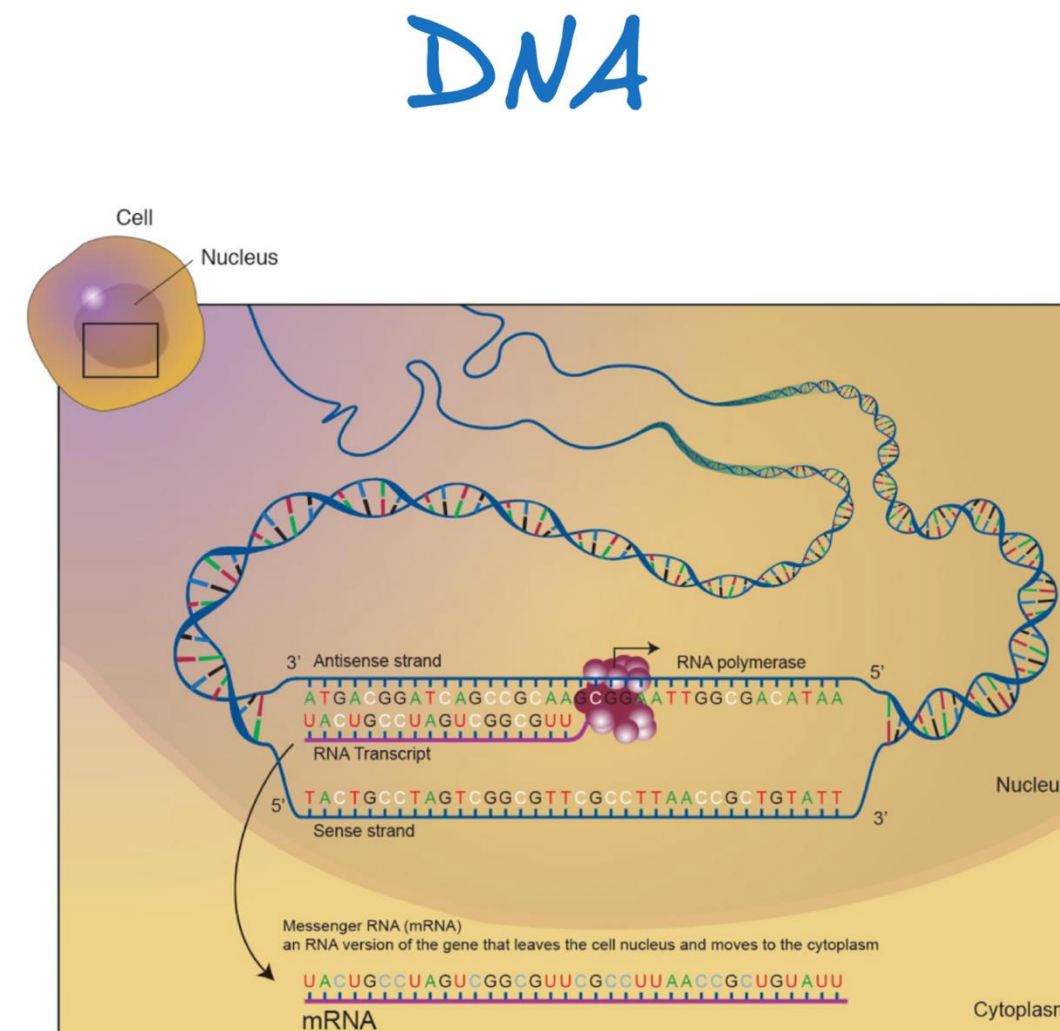
Concept Review



Why AI for Biological Sciences?



Systematic!!



Unsystematic...

What does mean “Unsystematic”?

DNA = kernel + interpreter + PYTHON + library +....

only with A, C, G, T

ATGACGGTCAGCCGCAACGGTACTTAAGATTTCGCC....

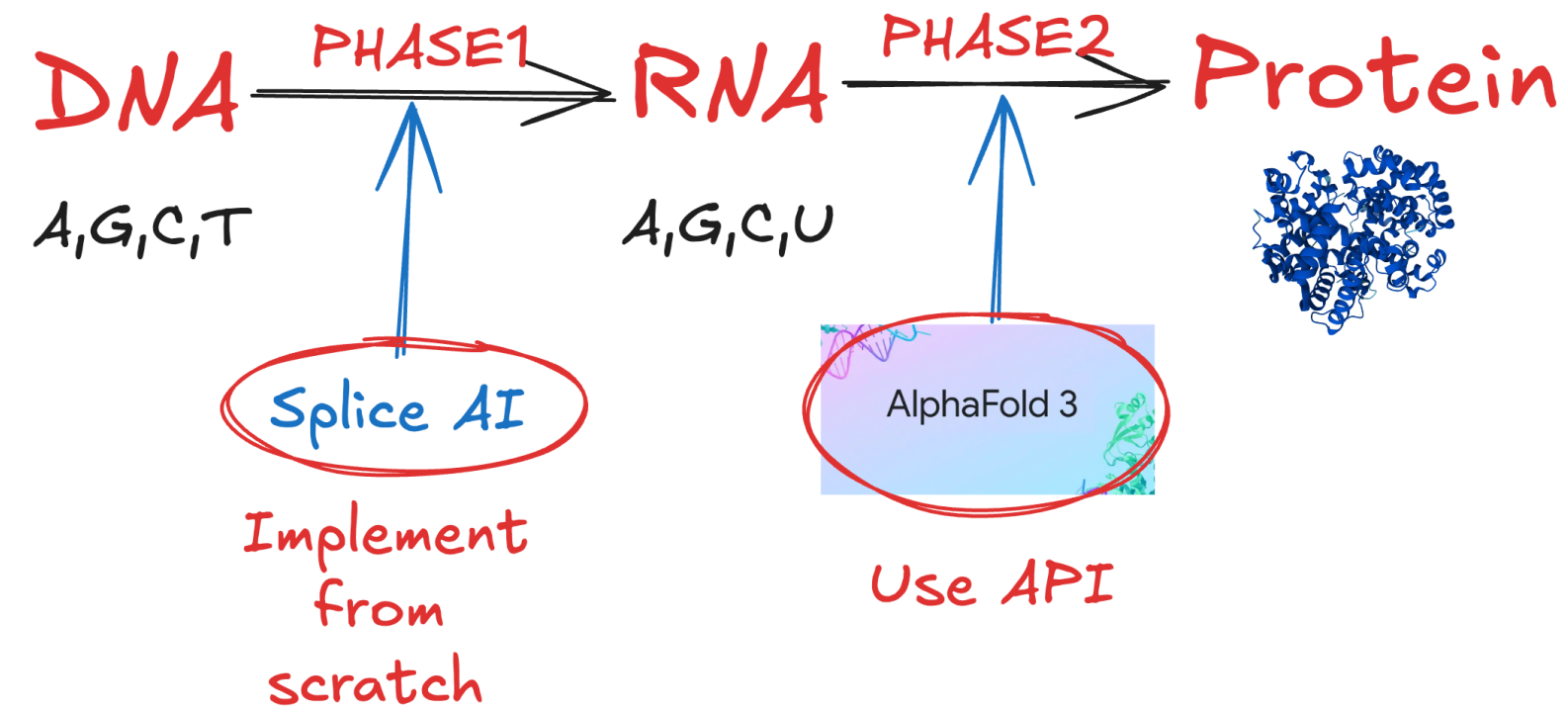
3.2billion characters!!...

32억글자!!!...

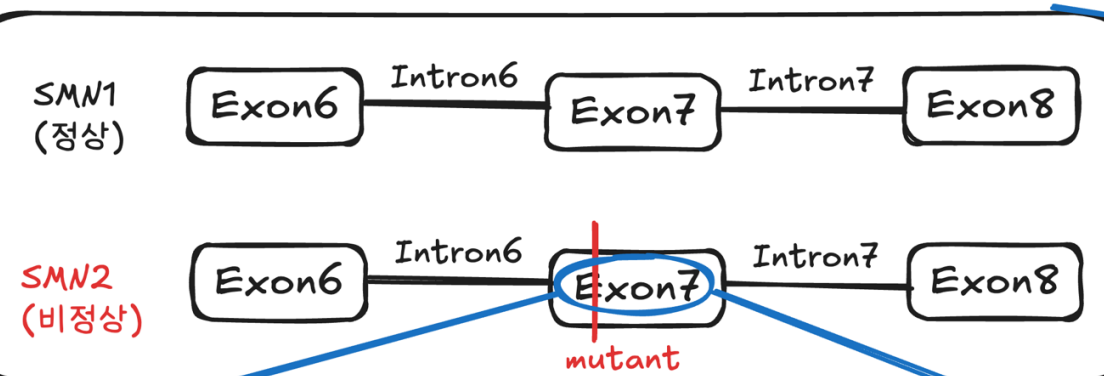
Until today, human can't interpret DNA

2. Splice AI Training Result

Splice AI



2. Manipulate DNA



ATAATTCCCCACCACCTCCCATATGTCCAGATTCTCTTGATGATGCTGATGCT
TTGGGAGTATGTTAATTTTCATGGTACATGAGTGGCTATCATACTGGCTAT
TATATG

User가
마음대로 조작 가능!

Next

Splice AI

3. Mature mRNA

SMN1(정상)



SMN2(비정상)



No Exon7!

Make Protein

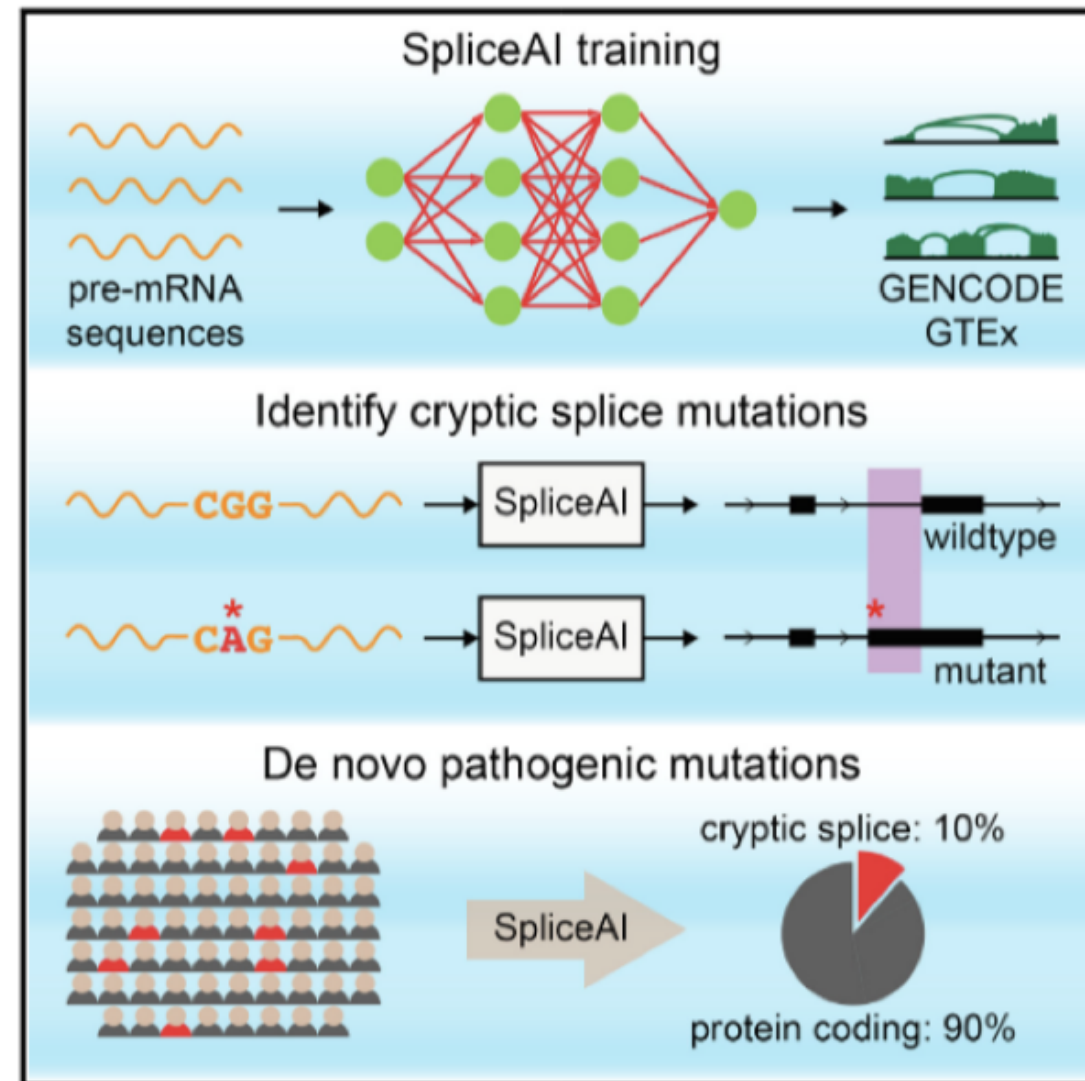
Splice AI

Article

Cell

Predicting Splicing from Primary Sequence with Deep Learning

Graphical Abstract



Authors

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Sofia Kyriazopoulou Panagiotopoulou,
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Correspondence

kfarh@illumina.com

In Brief

A deep neural network precisely models mRNA splicing from a genomic sequence and accurately predicts noncoding cryptic splice mutations in patients with rare genetic diseases.

Highlights

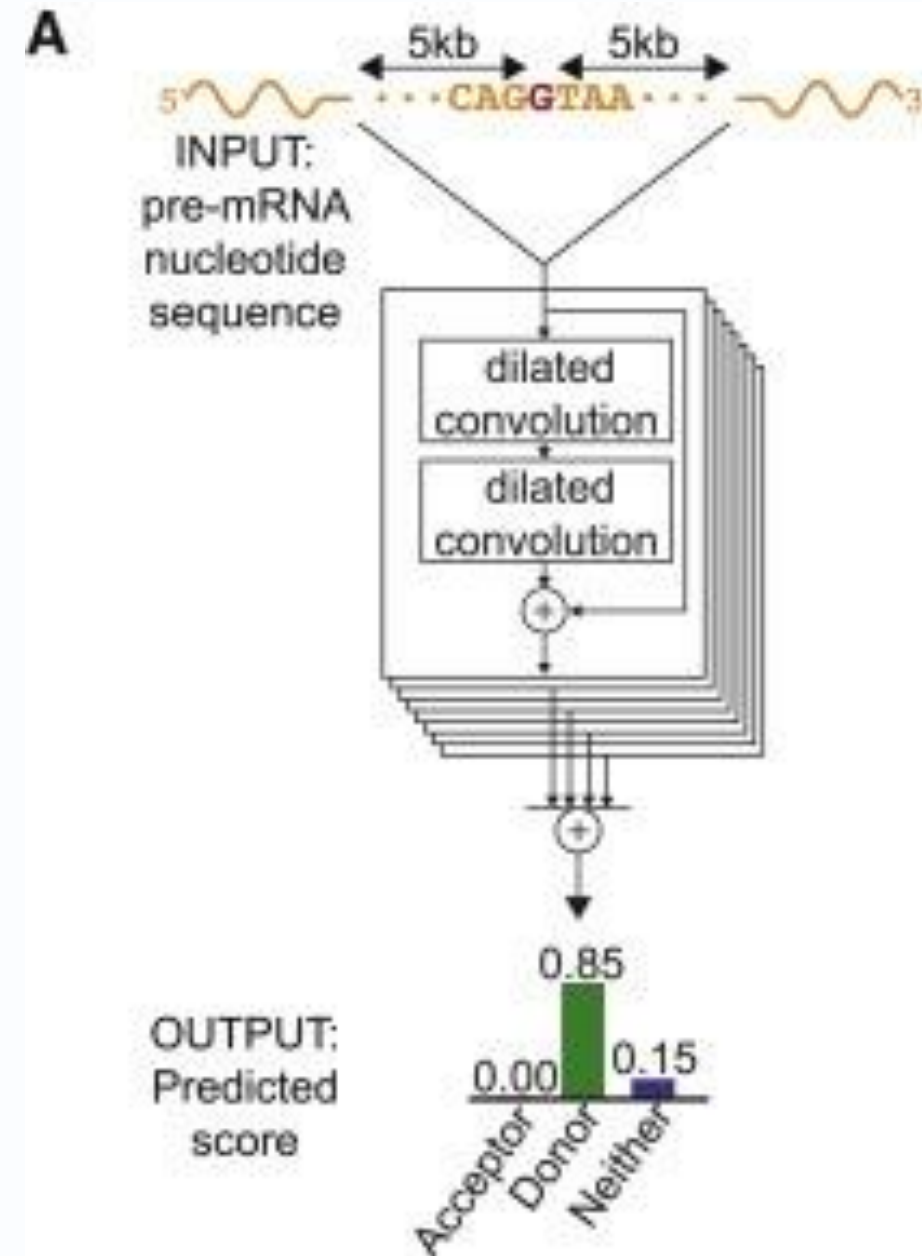
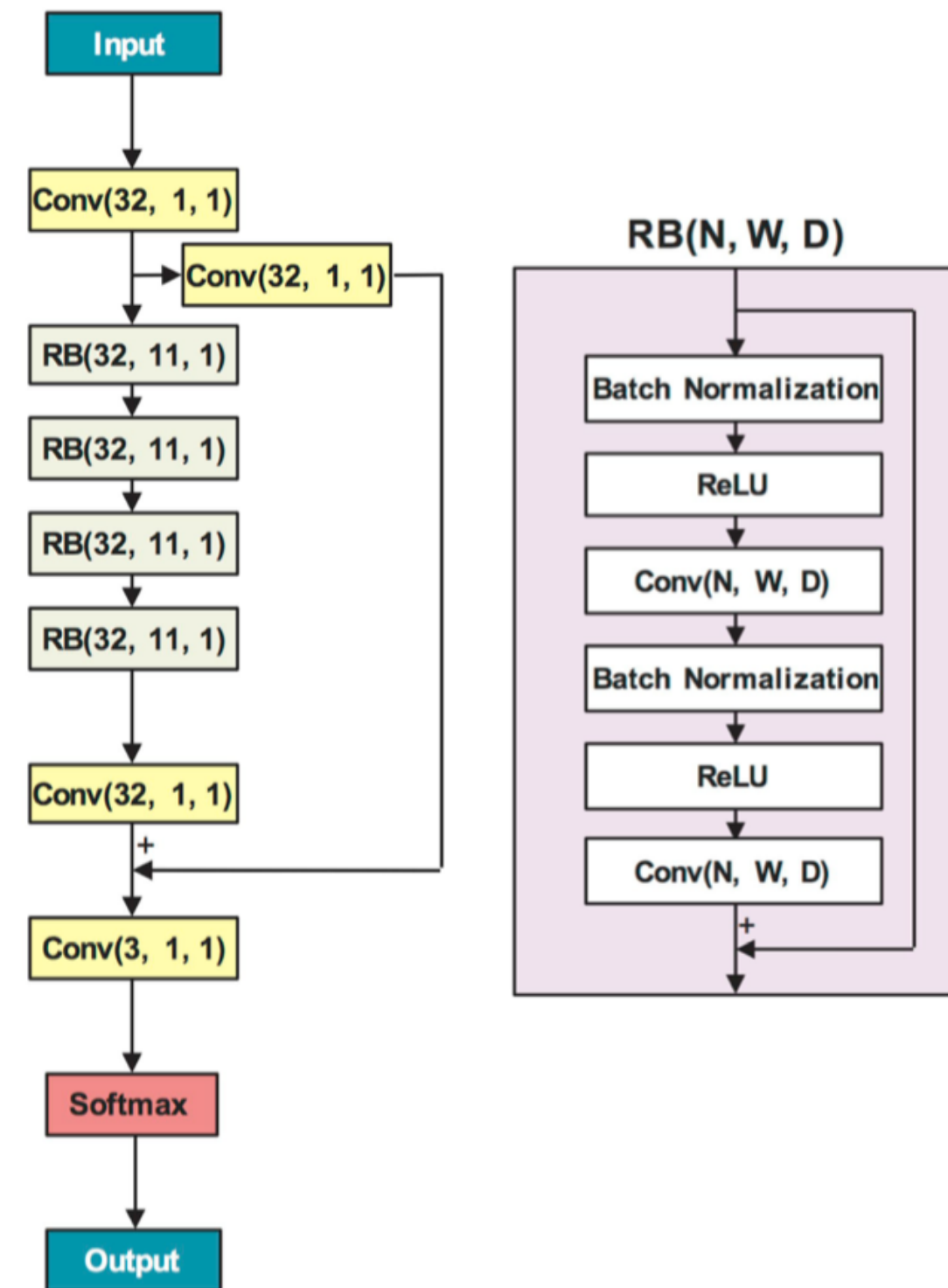
- SpliceAI, a 32-layer deep neural network, predicts splicing from a pre-mRNA sequence
- 75% of predicted cryptic splice variants validate on RNA-seq
- Cryptic splicing may yield ~10% of pathogenic variants in neurodevelopmental disorders
- Cryptic splice variants frequently give rise to alternative splicing



Jaganathan et al., 2019, Cell 176, 535–548
January 24, 2019 © 2018 Elsevier Inc.
<https://doi.org/10.1016/j.cell.2018.12.015>

Splice AI Model Structure

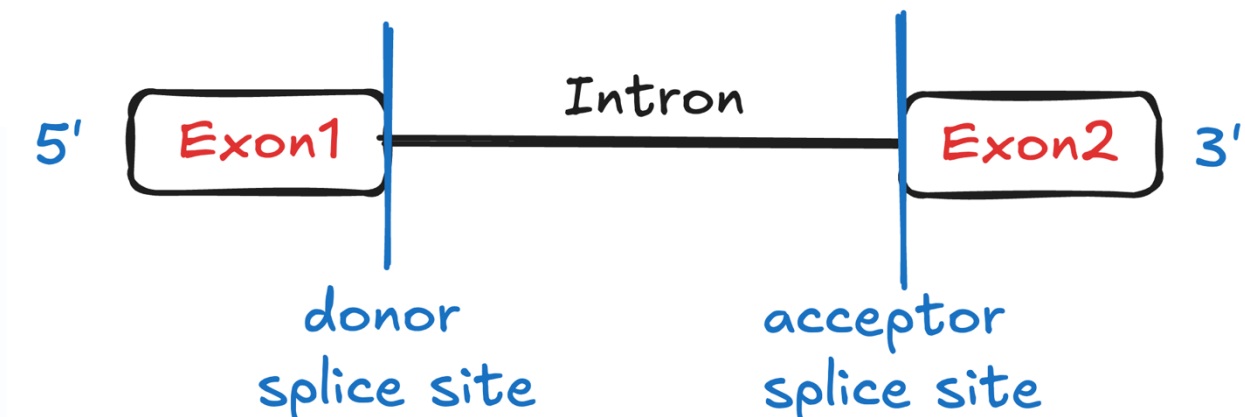
SpliceAI-80nt



- **Input:** one-hot encoded primary sequence

		A	T	G	G	C	T	C	A	T
A	[1,0,0,0]	1	0	0	0	0	0	0	1	0
C	[0,1,0,0]	0	0	0	0	1	0	1	0	0
G	[0,0,1,0]	0	0	1	1	0	0	0	0	0
T	[0,0,0,1]	0	1	0	0	0	1	0	0	1

- **Output:** donor / acceptor / neither (3 classes)



Studied SpliceAI paper at SNU Bio

교수

백대현 BAEK, DAEHYUN

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Lab Tel. : 02-880-4265
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Email : baek@snu.ac.kr



Took 1st place at SNU

성 적 증 명 서

성 명 : 강민준
생년월일 : 2004. 6. 18.
학 번 : 2025- [REDACTED]

과 정 : 국내교환
수학대학 : 자연과학대학 생명과학부
수학기간 : 2025. 3. 3. ~ 2025. 6. 14., 2025. 9. 1. ~ 2025. 12. 12.

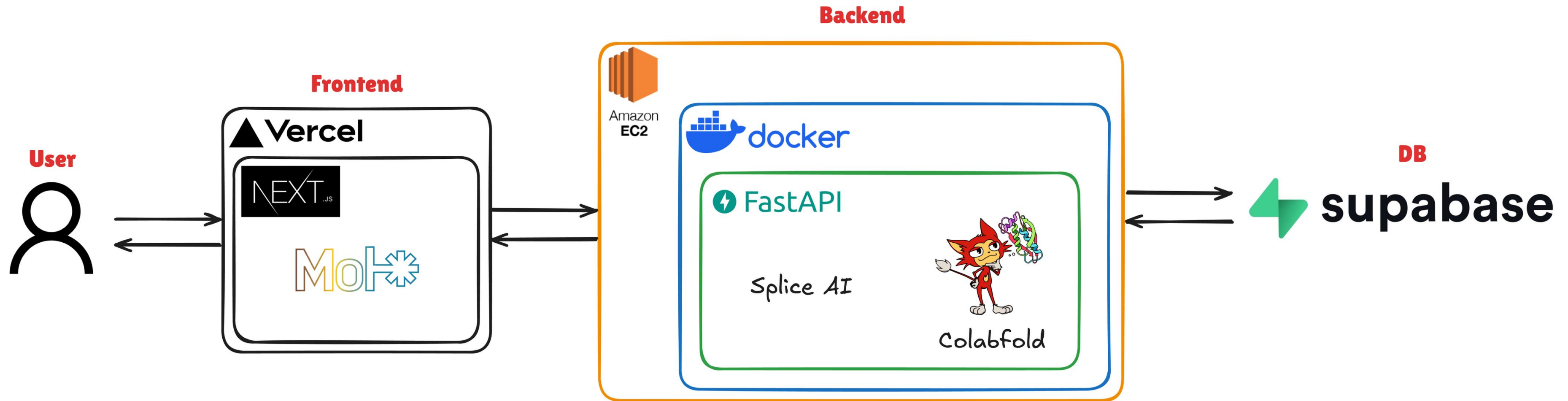
취 득 학 점

학 기	구분	교 과 목 명	학점	성적	학 기	구분	교 과 목 명
2025. 1.	교양	기초생물학 1	1	S			
2025. 1.	교양	생물학실험	1	A+			
2025. 1.	일선	인공지능 개론	3	S			
2025. 1.	대학원	영어논문작성법: 이공계열	1	S			
		학점계 : 6					
		평점평균 : 4.30					
2025. 2.	전선	생명과학을 위한 인공지능	4	A+			
		학점계 : 4					
		평점평균 : 4.30					
2025. 2.	전선	생명과학을 위한 인공지능	4	A+			
		평점평균 : 4.30					
		백점환산점수 : 100.00					

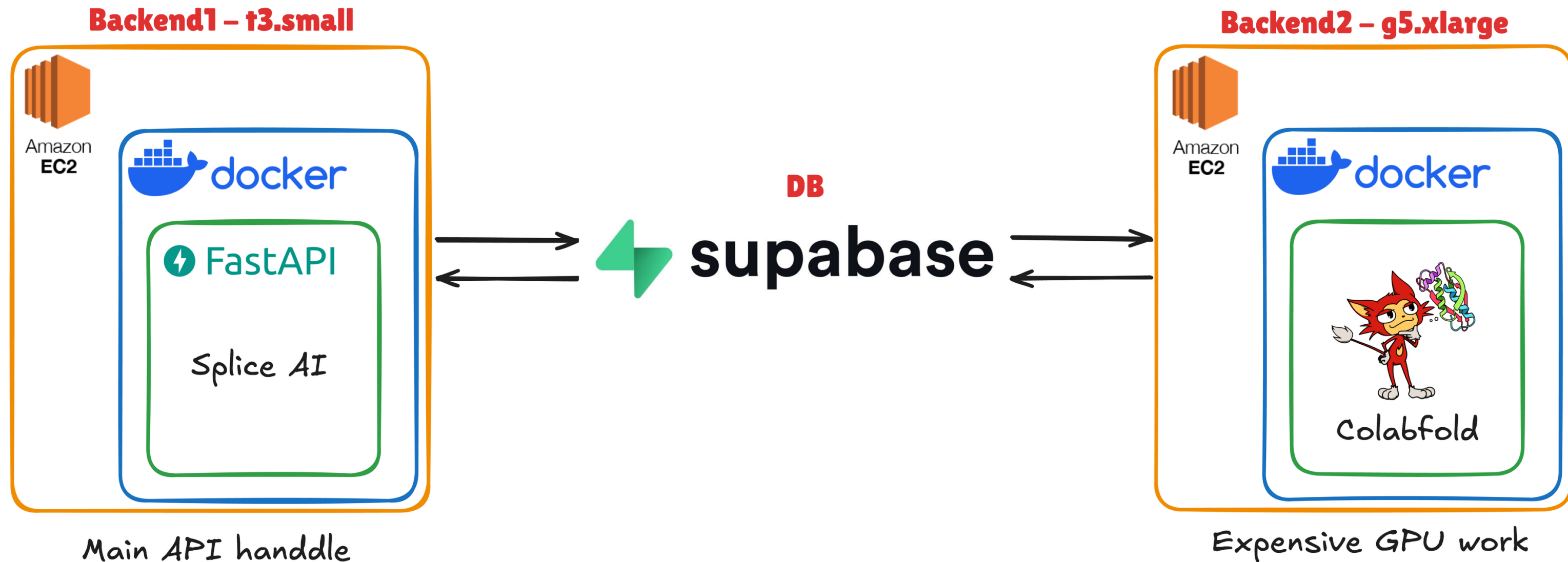
2025. 2. | 전선 | 생명과학을 위한 인공지능 | 4 | A+

3. System Architecture

System Architecture



Backend details – Asynchronous Operation



t3.small

Family: **t3** 2 vCPU 2 GiB Memory Current generation: true
On-Demand RHEL base pricing: 0.0496 USD per Hour
On-Demand Windows base pricing: 0.0392 USD per Hour
On-Demand SUSE base pricing: 0.0518 USD per Hour
On-Demand Linux base pricing: 0.0208 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0243 USD per Hour

g5.xlarge

Family: **g5** 4 vCPU 16 GiB Memory Current generation: true
On-Demand Ubuntu Pro base pricing: 1.013 USD per Hour
On-Demand Windows base pricing: 1.19 USD per Hour
On-Demand RHEL base pricing: 1.0636 USD per Hour
On-Demand SUSE base pricing: 1.0623 USD per Hour
On-Demand Linux base pricing: 1.006 USD per Hour

Backend details – Asynchronous Operation

Backend1 – t3.small



Amazon
EC2

- **CPU only**
- **API handle**



supabase

DB



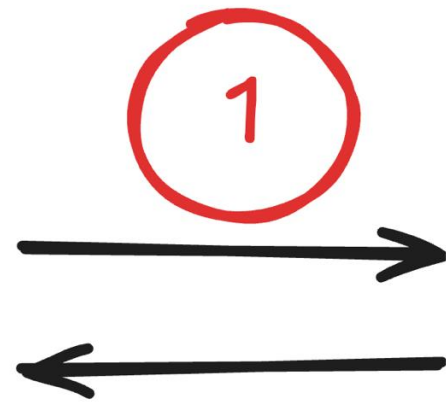
Amazon
EC2

Backend2 – g5.xlarge

- **on / off GPU worker!!**

Backend details – STEP1

Backend1 – t3.small



DB



supabase

Backend2 – g5.xlarge



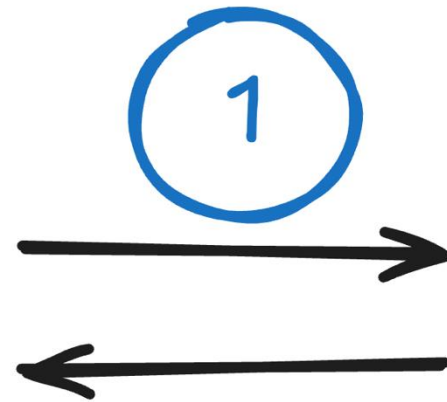
1. GPU연산이 필요 없는 STEP1~3에 대해 API handle (Splice AI 포함)
& STEP3에서 나온 결과인 '단백질 서열'을 DB에 업로드

Backend details – STEP2

Backend1 – t3.small



Amazon
EC2



DB



supabase

Backend2 – g5.xlarge

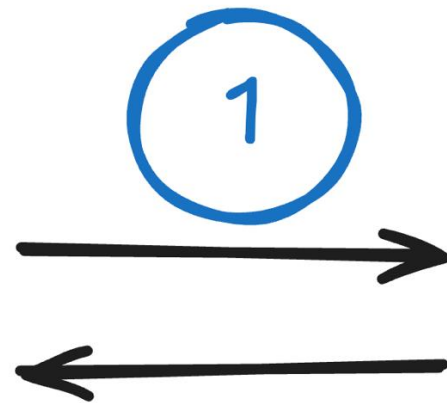


Amazon
EC2

2. DB에 올라온 '단백질 서열'을 읽어서 구조 예측(Colabfold!!)
& 예측한 3D 단백질 구조 파일 만들어서 다시 DB에 저장

Backend details – STEP3

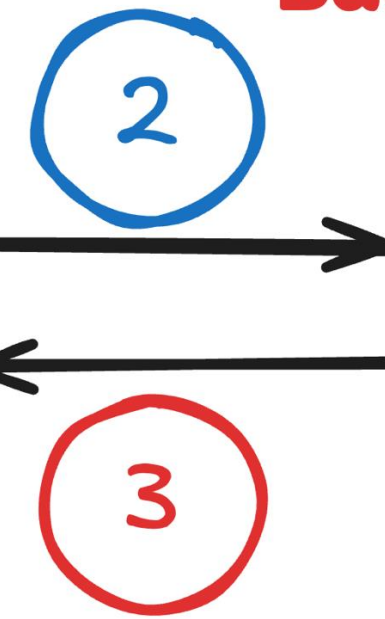
Backend1 – t3.small



DB

supabase

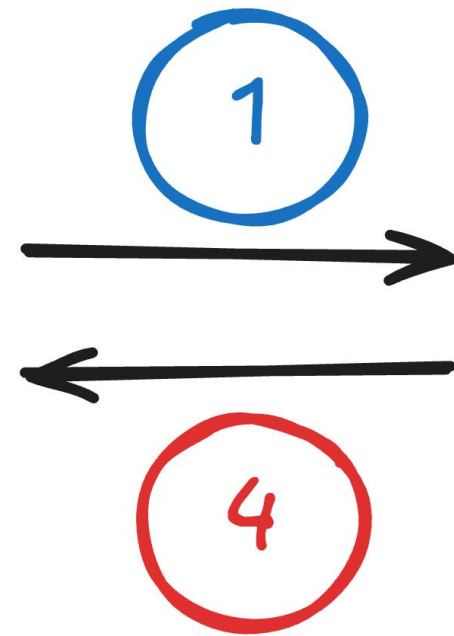
Backend2 – g5.xlarge



3. & 예측한 3D 단백질 구조 파일 만들어서 다시 DB에 저장

Backend details – STEP4

Backend1 – t3.small

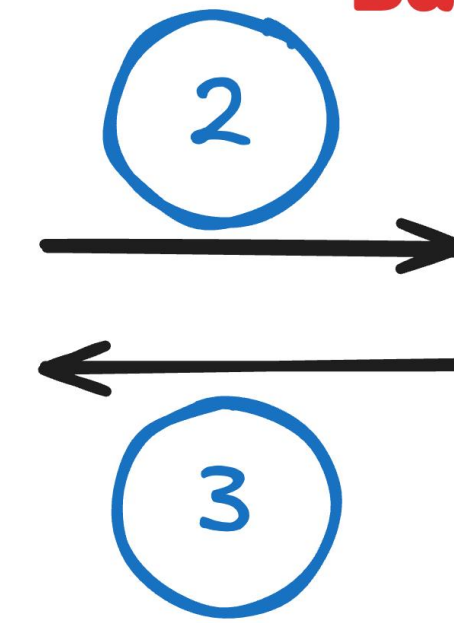


DB



supabase

Backend2 – g5.xlarge



4. API handling하는 main 백엔드 서버가 단백질 구조 파일을 받아온 후,
프론트엔드에 단백질 구조 파일 반환

AlphaFold -> ColabFold

Brief Communication | [Open access](#) | Published: 30 May 2022

ColabFold: making protein folding accessible to all

[Milot Mirdita](#) ✉, [Konstantin Schütze](#), [Yoshitaka Moriwaki](#), [Lim Heo](#), [Sergey Ovchinnikov](#) ✉ & [Martin Steinegger](#) ✉

[Nature Methods](#) **19**, 679–682 (2022) | [Cite this article](#)

301k Accesses | **9013** Citations | **376** Altmetric | [Metrics](#)



science
자 연 과 학 대 학 · 생 명 과 학 부

부교수

마틴 슈타이네거 MARTIN STEINEGGER

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Home

Download Structure

Source	PDB
PDB Id(s)	1tqn

Options

Apply

Download Density

Download File

Open Files

Load Trajectory

Download

Load Genome 3D (G3D)

Download Tunnels

Zenodo Import

Remote States

Zika+EM

P-450 Superposition & Validation

NPC

1RB8 Assembly Symmetry

Wed, 08 Apr 2026 02:28:52 GMT

Sequence of SARS-COV-2... Chain 1: Lipid bilayer 1

[1/1] 1/1/2022, 4:19:54 AM

13:57:39 Created Spacefill in 2ms.

13:57:39 Created S-protein trimers (glycosylated) in 0ms.

13:57:39 Created Spacefill in 2ms.

Structure Tools

Structure

SARS-COV-2_VIRION

Type Model

Nothing Focused

Measurements

Add

Quick Styles

Apply Representation

Default Cartoon Spacefill Surface

Apply Style

Default Illustrative

Components SARS-COV-2_VIRION

Preset Add

Lipid bilayer Spacefill

M-protein dimers Spacefill

E-protein pentamers

S-protein trimers (glycos...

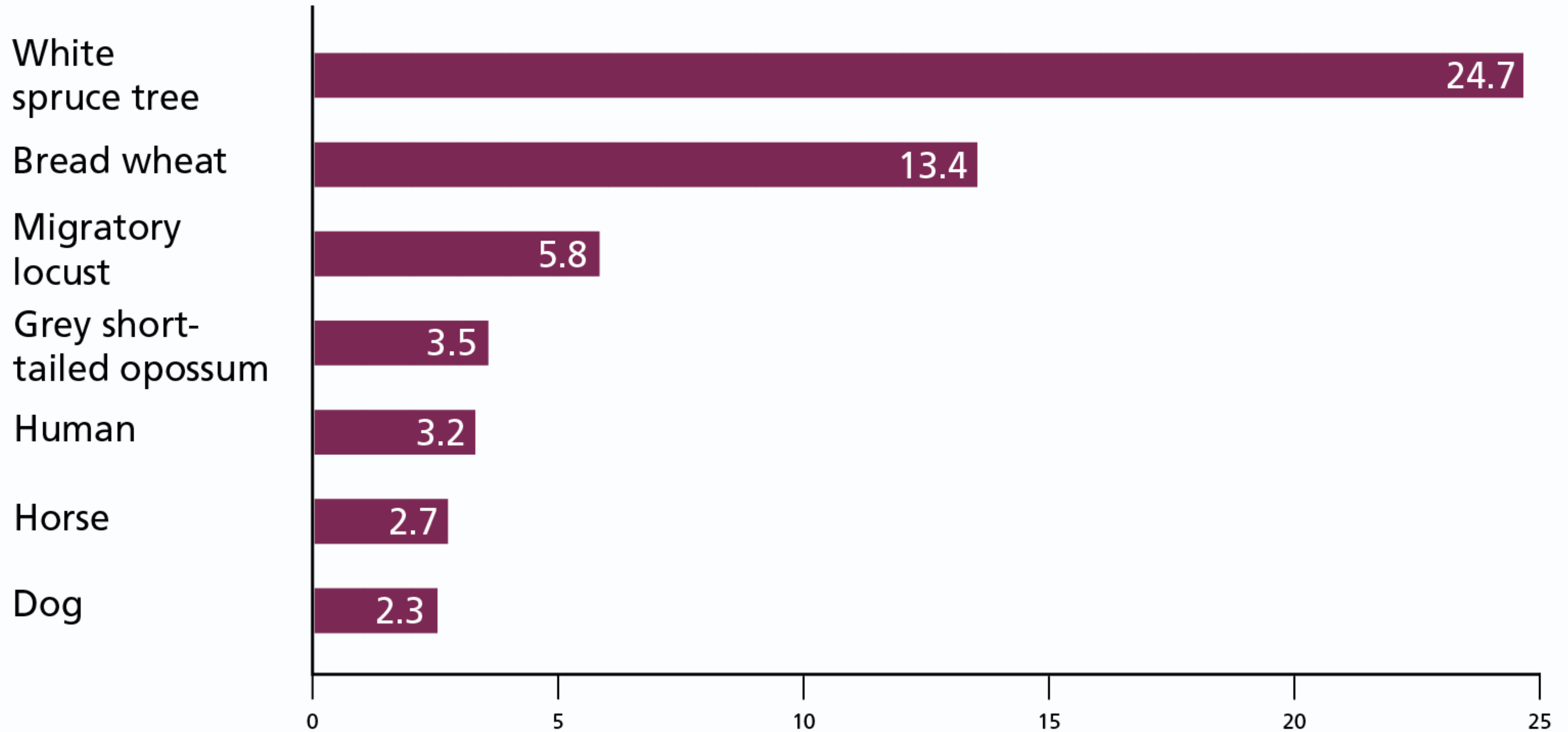
Export Models

Export Animation

Export Geometry

4.DB Construction

Human Genome Size – 3.2billion!...



DB upload pipeline

Raw data

FASTA file



3.2billion length(32억글자!..)

GTF file

gencode_v19_annotation_gtf[gencode_v19_annotation_gtf['gene_name']=='TERT']													
seqname	source	feature	start	end	score	strand	frame	gene_id	transcript_id	gene_type	gene_status	gene	
696990	chr5	HAVANA	gene	1253262	1295184	NaN	-	0	ENSG00000164362.14	ENST00000164362.14	protein_coding	KNOWN	
696991	chr5	HAVANA	transcript	1253262	1283125	NaN	-	0	ENSG00000164362.14	ENST00000484238.2	protein_coding	KNOWN	
696992	chr5	HAVANA	exon	1282544	1283125	NaN	-	0	ENSG00000164362.14	ENST00000484238.2	protein_coding	KNOWN	
696993	chr5	HAVANA	exon	1280273	1280453	NaN	-	0	ENSG00000164362.14	ENST00000484238.2	protein_coding	KNOWN	
696994	chr5	HAVANA	exon	1279406	1279585	NaN	-	0	ENSG00000164362.14	ENST00000484238.2	protein_coding	KNOWN	
696995	chr5	HAVANA	exon	1278756	1278911	NaN	-	0	ENSG00000164362.14	ENST00000484238.2	protein_coding	KNOWN	
696996	chr5	HAVANA	exon	1268635	1268748	NaN	-	0	ENSG00000164362.14	ENST00000484238.2	protein_coding	KNOWN	
696997	chr5	HAVANA	exon	1266579	1266650	NaN	-	0	ENSG00000164362.14	ENST00000484238.2	protein_coding	KNOWN	
696998	chr5	HAVANA	exon	1260589	1260715	NaN	-	0	ENSG00000164362.14	ENST00000484238.2	protein_coding	KNOWN	
696999	chr5	HAVANA	exon	1258713	1258774	NaN	-	0	ENSG00000164362.14	ENST00000484238.2	protein_coding	KNOWN	
697000	chr5	HAVANA	exon	1255402	1255526	NaN	-	0	ENSG00000164362.14	ENST00000484238.2	protein_coding	KNOWN	

table data

INPUT
specific gene that we want

UPLOADER



python™

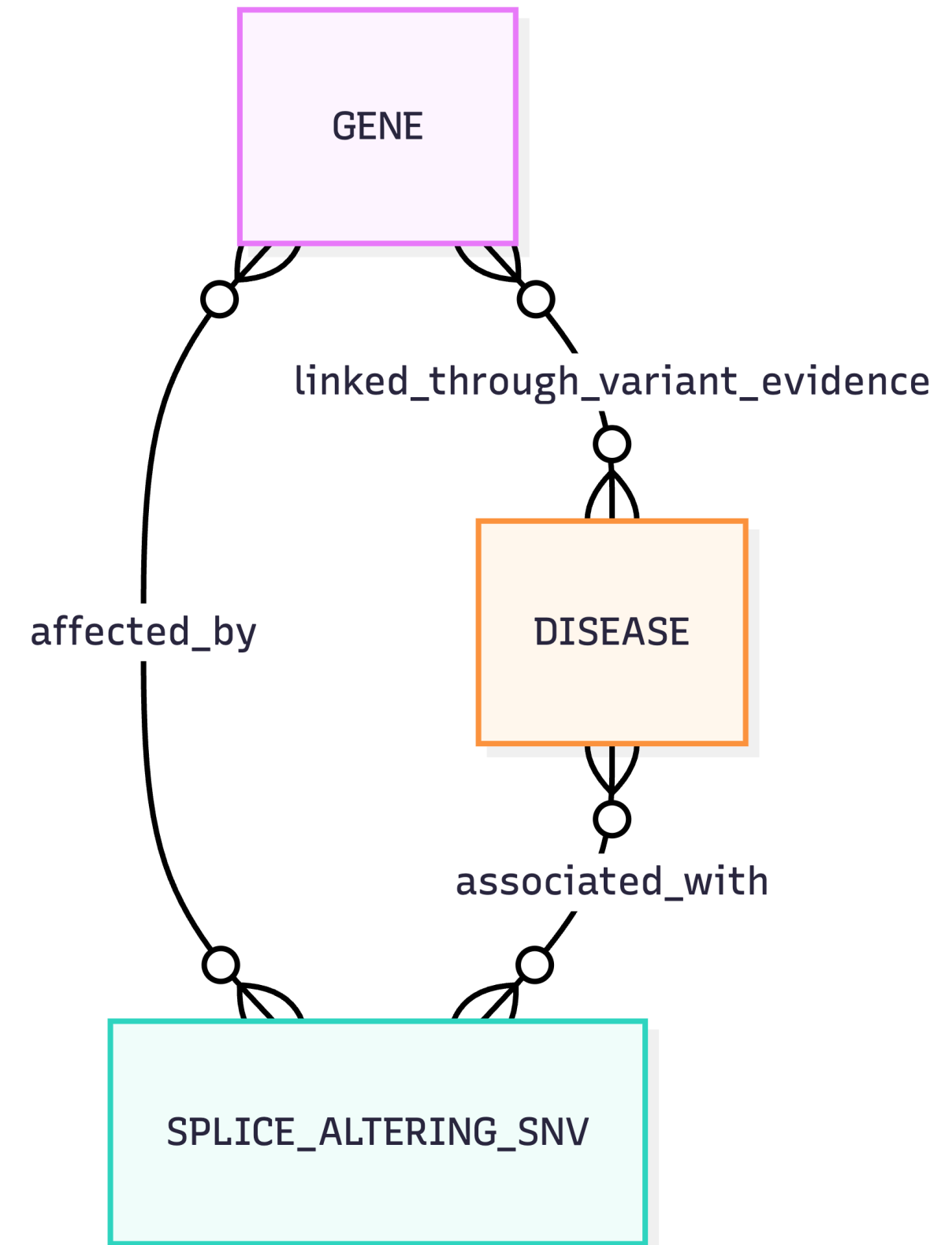
DB



supabase

Gene–Disease–SNV Many-to-Many DB

- Variant itself is stored independently
- biological meaning is attached through gene/transcript/disease evidence tables.



ClinVar BRCA1 Gene Variation Example



National Library of Medicine
National Center for Biotechnology Information

Log in

ClinVar Home

About ▾

Understand the Data ▾

Access the Data ▾

Submit Data ▾

Search 🔍

ClinVar

15897 search results for **BRCA1**

Classification type

☐ Germline (13,858)

☐ Somatic (11)

Germline classification

☐ Conflicting classifications (2,912)

☐ Benign (832)

☐ Likely benign (2,908)

☐ Uncertain significance (2,567)

☐ Likely pathogenic (556)

☐ Pathogenic (4,352)

Types of conflicts

☐ P/LP vs LB/B (6)

☐ P/LP vs VUS (107)

☐ VUS vs LB/B (2,805)

Molecular consequence

☐ Frameshift (2,794)

Graphical view of search results ▲

GRCh37

Gene

Pathogenic

Likely pathogenic

Uncertain significance

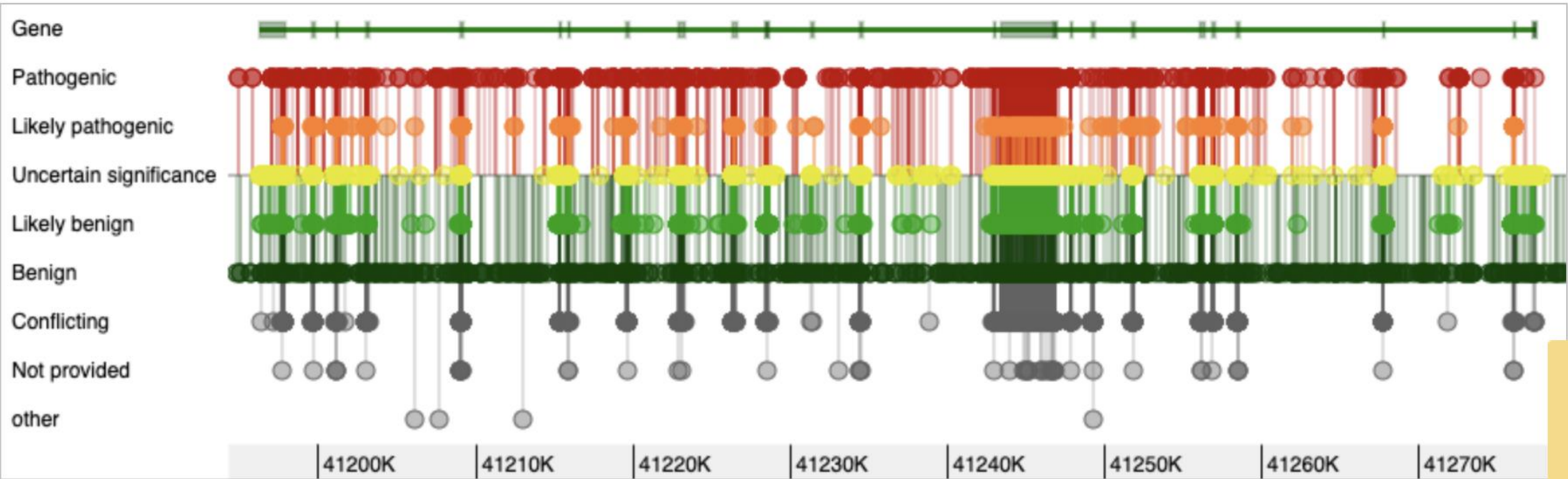
Likely benign

Benign

Conflicting

Not provided

other



41200K 41210K 41220K 41230K 41240K 41250K 41260K 41270K

Search results

Display options ▾

Sort by Location ▾

Download ▾

Items: 1 to 100 of 15897

<< First

< Prev

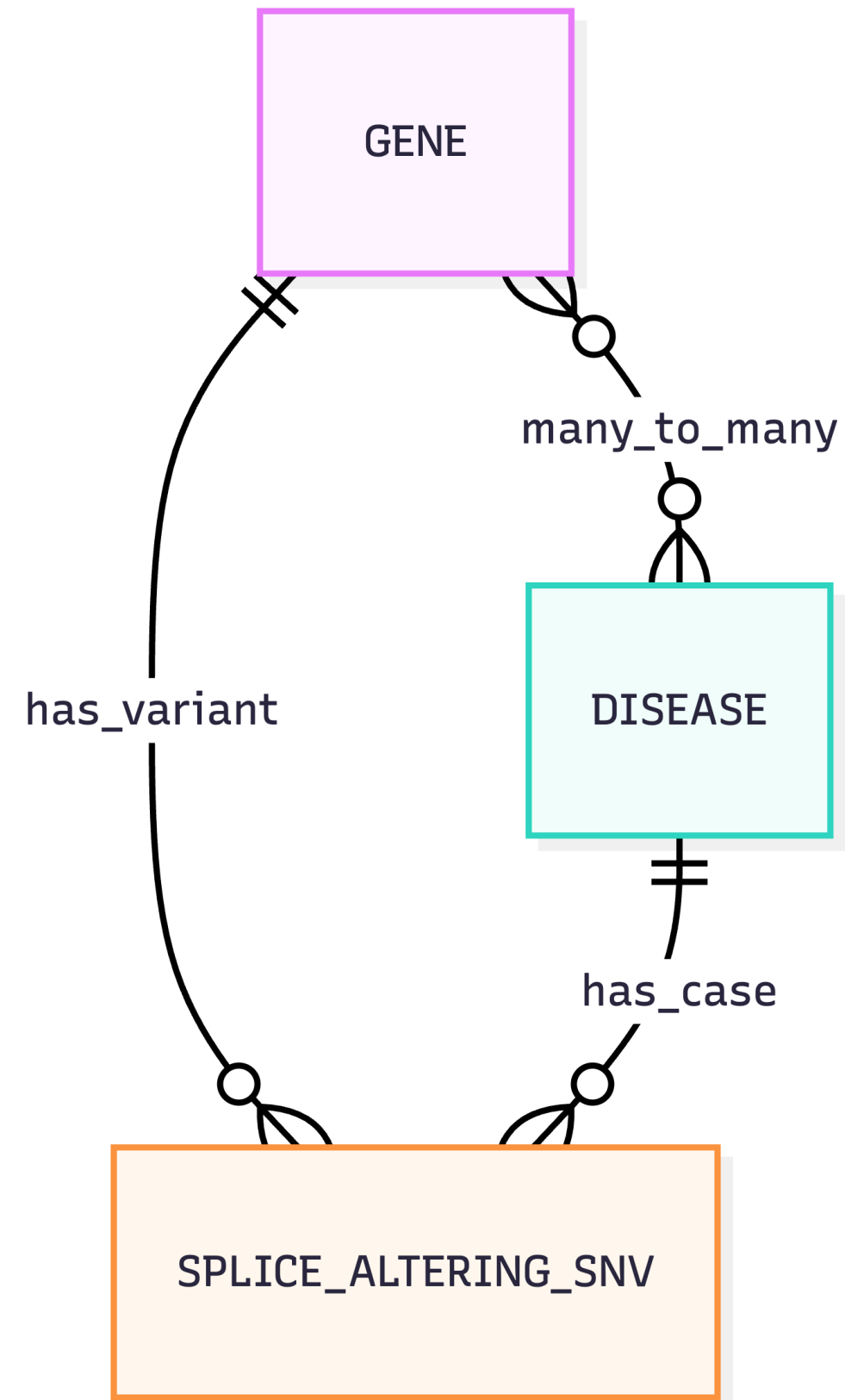
Page 1 of 159

Next >

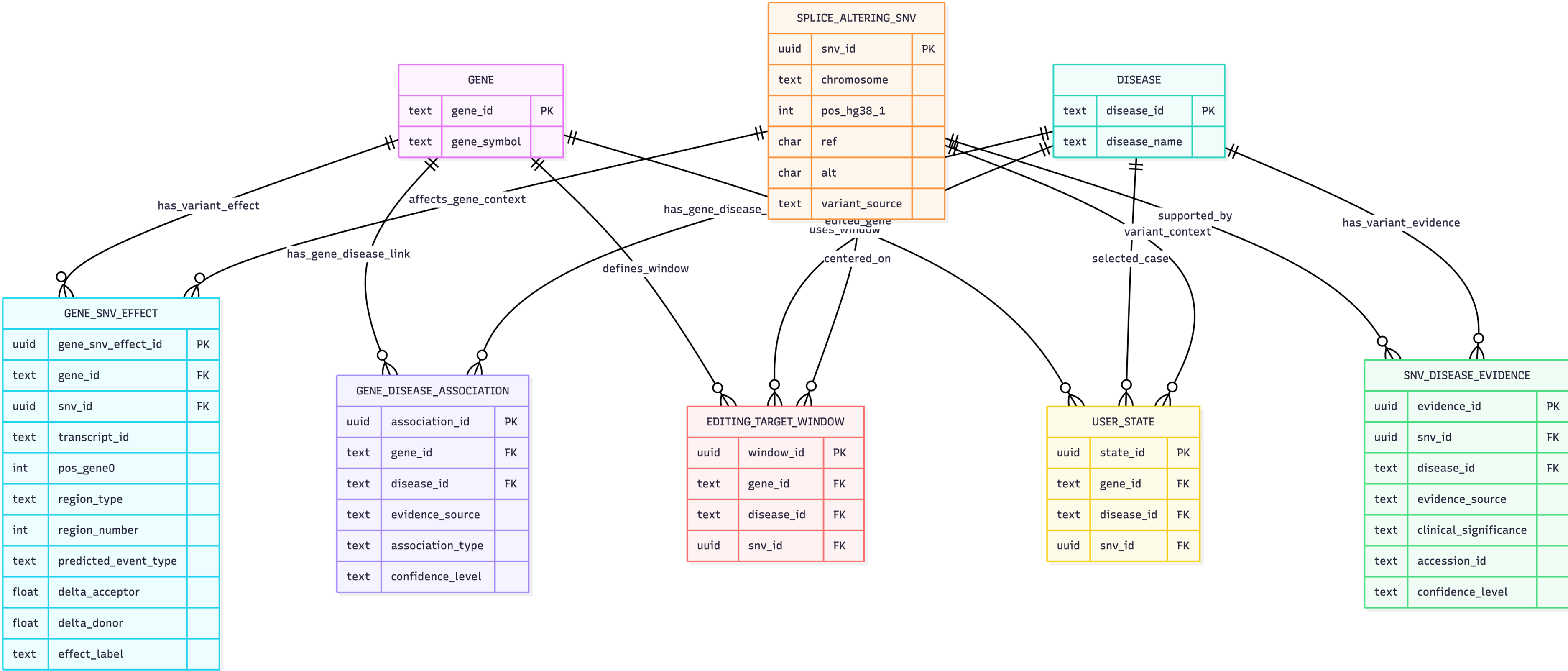
Last >>

Feedback

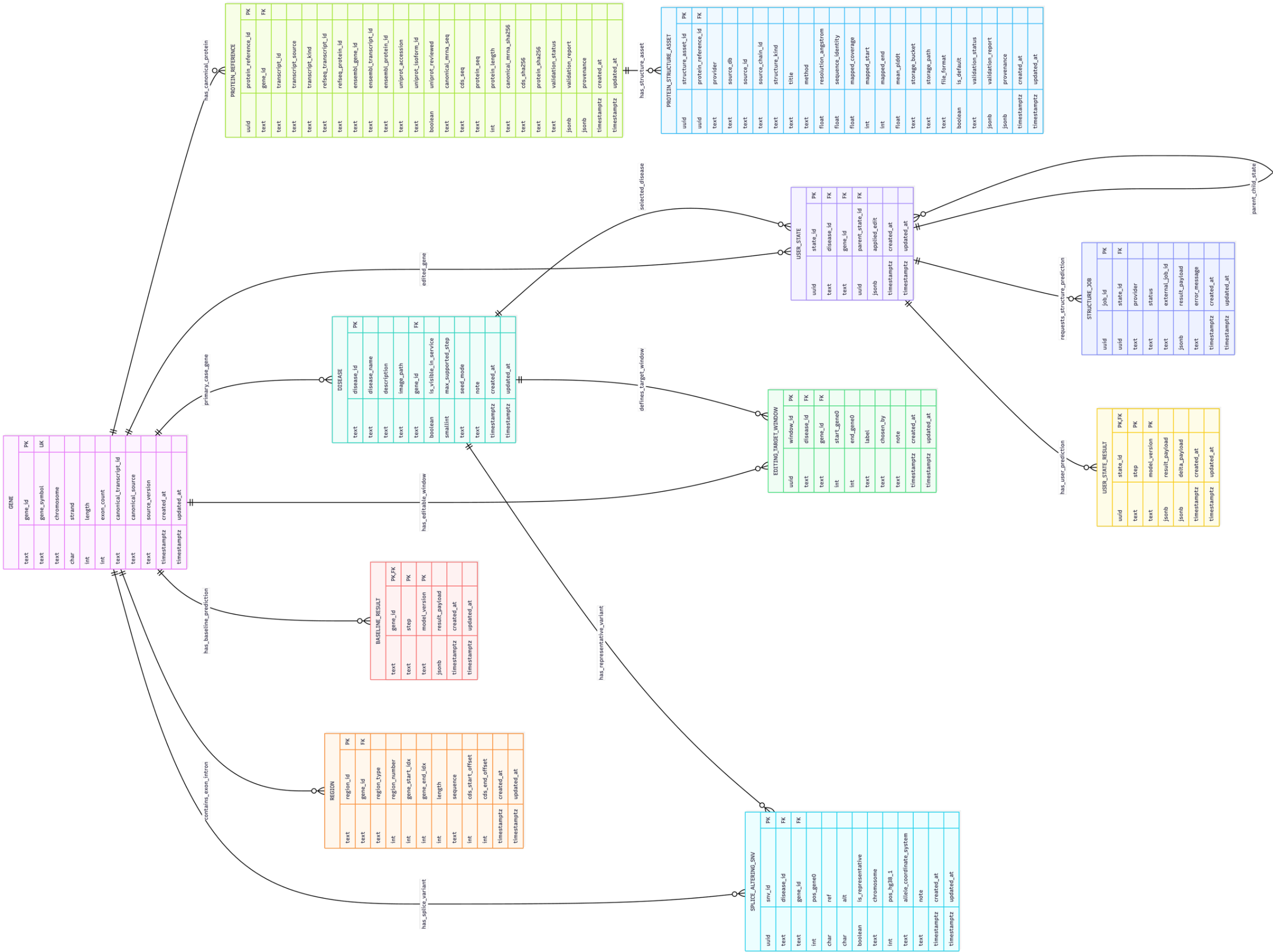
Gene–Disease–SNV Real Implementation



Gene–Disease–SNV Many-to-Many Expansion



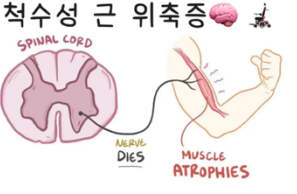
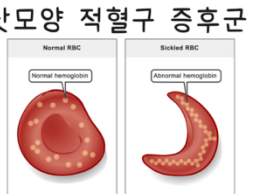
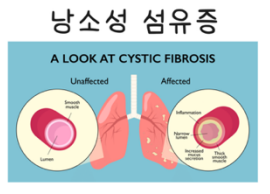
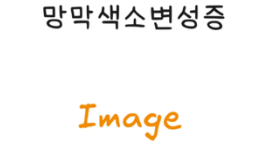
Whole Service ER-Diagram in SUPABASE



5. Progress & Demo

User Interface Review

1. Select Mutant

 척수성 근 위축증 SPINAL CORD MUSCLE ATROPHIES	 낫모양 적혈구 증후군 Normal RBC Sickled RBC	 낭소성 섬유증 A LOOK AT CYSTIC FIBROSIS Unaffected Affected
Spinal Muscular Atrophy	sickle cell anemia	Cystic Fibrosis
 뒤센 근이영양증 Image	 가족성 고콜레스테롤혈증 Image	 망막색소변성증 Image
Duchenne Muscular Dystrophy	Familial Hypercholesterolemia	Retinitis Pigmentosa

2. Manipulate DNA

SMN1 Exon6 Intron6 Exon7 Intron7 Exon8

SMN2 Exon6 Intron6 ~~Exon7~~ Intron7 Exon8

mutant

Chromosome5 SMN1, SMN2

User가 마음대로 조작 가능!

ATAATTCCTCCACACCTCCCATATGTCCAGATTCTCTTGATGATGCTGATGCT
TTGGGAAGTATGTTAATTTTCATGGTACATGAGTGGCTATCATACTGGCTAT
TATATG

Next

3. Mature mRNA

SMN1 (정상)

Exon1 Exon2 Exon3 Exon4 Exon5 Exon6 Exon7 Exon8

SMN2 (비정상)

Exon1 Exon2 Exon3 Exon4 Exon5 Exon6 Exon8

No Exon7 !

Make Protein

4. Protein structure

SMN1 (정상)

SMN2 (비정상)





SPLICE PLAYGROUND

Visualize Splicing Into Structure

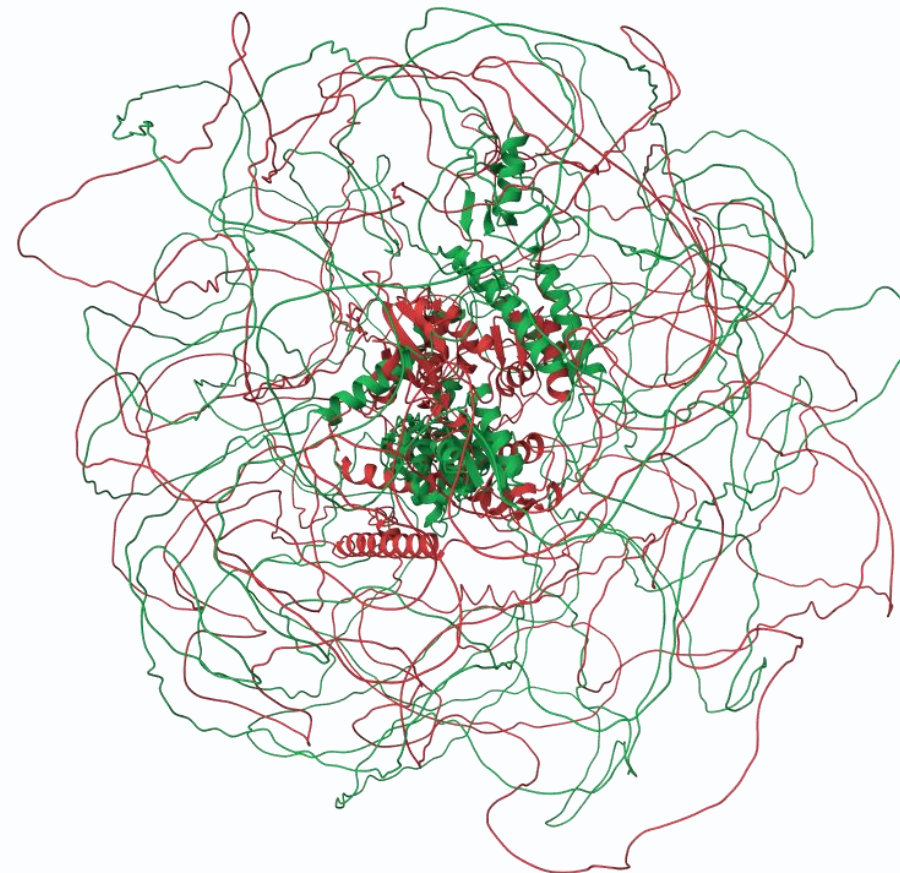
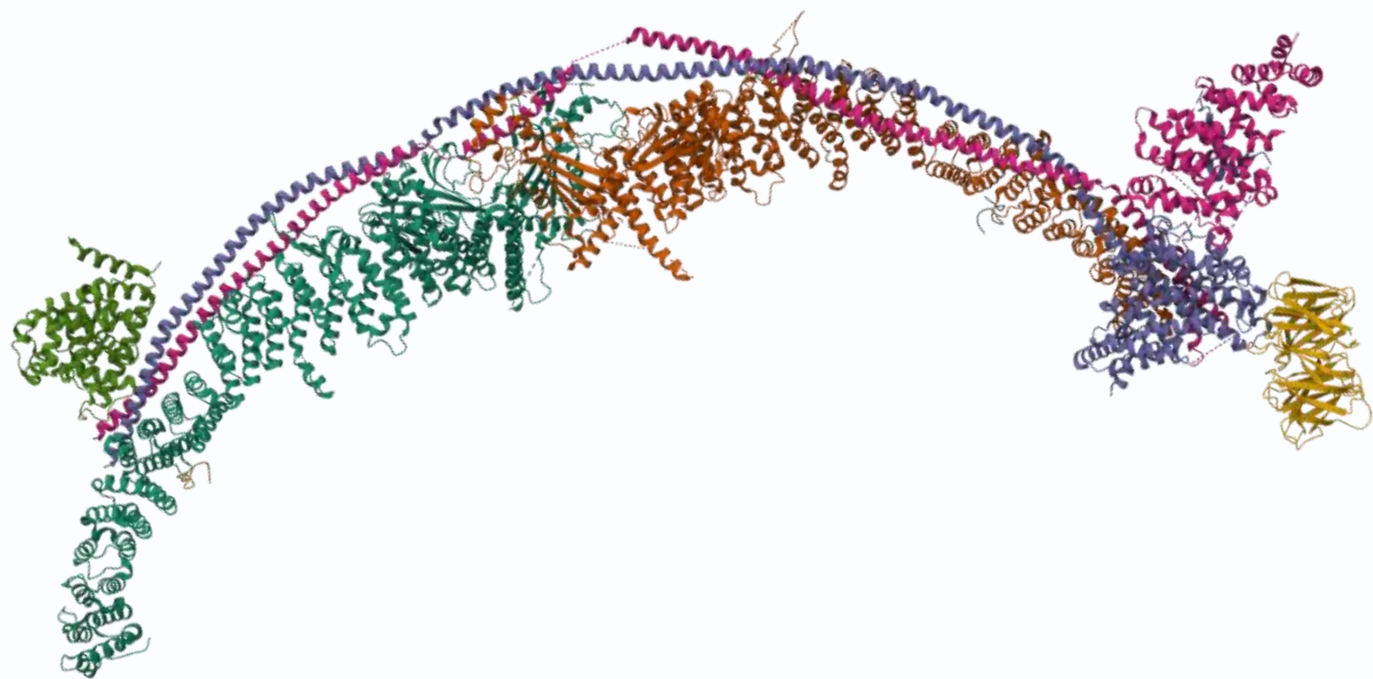
DNA editing, transcript consequences, and protein structure prediction in one
continuous flow

Start

6. Future Work & Mentoring

STEP5!! – Molecular Interaction Simulation

- How does this produced protein(at STEP4) effect our body?
- Degraded?, Autoimmune disease?, Cell destruction?, only Malfunction?





대표이사 컴퓨터공학부 교수 (AI/생물정보학 경력 30년)

- ✓ SCI급 논문 200여편 (Nature Communications 등)
- ✓ 국내 저명 학회 및 위원회 주요 요직 역임



R&D 팀 (4명)

총괄 (생물정보학 박사)

- SCI급 논문 10여편 이상 게재
- 대형 제약사/화장품사 공동연구 수행 경험



플랫폼 팀 (4명)

총괄 (컴퓨터공학 석사)

- AI 암 진단 플랫폼 개발 경험
- 웹/인프라 구축 전문가



자문 네트워크

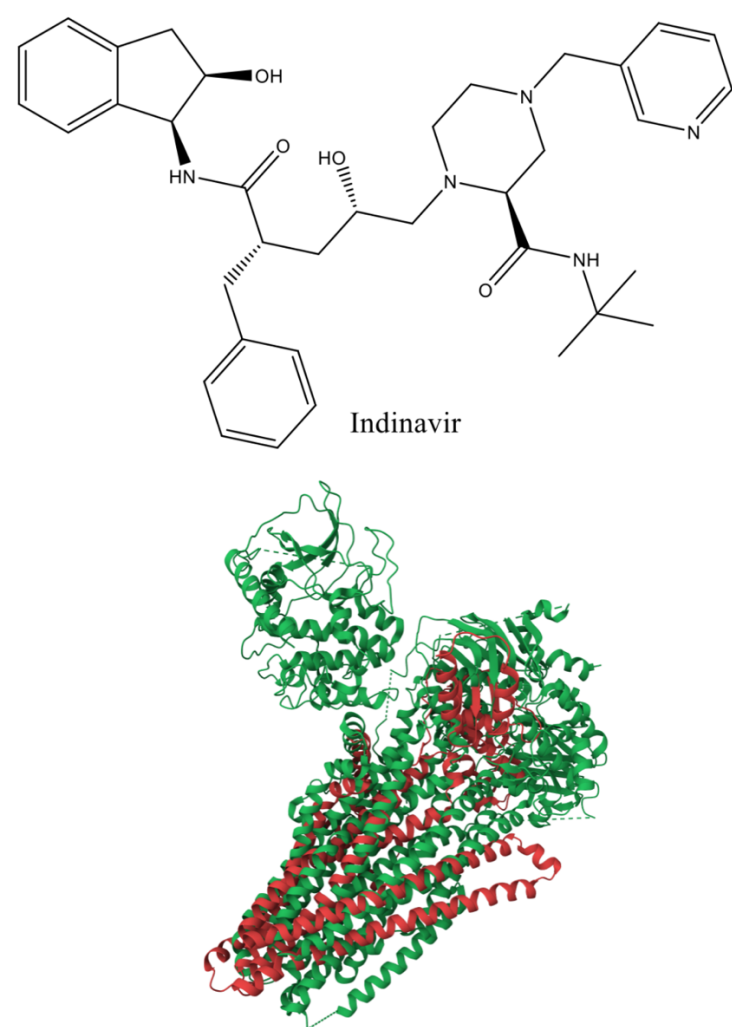
미국: Stanford, Princeton, Weill Cornell, Cedars Sinai Medical Center

국내: 서울대학교, 숙명여대, 동국대, 인하대, 경북대, 충북대, 명지대 등

실제 연구 및 개발에 직접 참여하는 유기적 자문 네트워크

- 신약개발 관련 연구 협력
- Agentic AI 등 자문
- AI 모델 개발 자문
- 플랫폼 개발 자문

Step5 Concept



Drug

Chemical
Language
Model

Protein

Protein
Language
Model

Fusion
Model

Affinity Score
 K_d , K_i , IC_{50} ...

Mentoring (at June 2nd)

- 진준우 멘토

- GPU서버를 켜다 켜다 할 수 있는 기능이 AWS의 Batch라는 서비스에 있다. 이를 활용하면 좋을 것 같다.
- 수많은 사람들에게 서비스를 한다고 가정했을 때 3D file 최적화가 필요해 보인다.

- 이승화 멘토

- 마찬가지로 AWS의 Batch라는 서비스를 활용해 보면 좋을 것 같다.

07

창업플랜 (Business Plan)

Splice Playground: 가상 유전체 실험 플랫폼의 상용화 전략

1. 회사 및 제품 목적

생물학 실험의 높은 진입장벽 해소

- ▲ 현실적 한계: 실험 장비, 시약, 랩 마우스 등 막대한 비용과 윤리적 심의 절차 필요.
- 🧠 AI 솔루션: 실제 생명체 조작 없이, SpliceAI와 ColabFold를 활용한 100% 가상(In-silico) 실험 환경 제공.
- ✅ 핵심 가치: 유전 변이의 결과를 3D 구조 수준에서 즉각 시각화하여 학습 효율 극대화.



VISION & MISSION

"생물학의 스크래치(Scratch)"

복잡한 코딩을 몰라도 Scratch에서 블록을 조립하듯, 분자생물학의 세부 기작을 몰라도 유전 변이를 조립하고 결과를 탐구할 수 있는 세상을 만듭니다.



실험의 민주화
누구나 안전하게 유전체 실험을 수행



융합형 인재 양성
AI와 바이오를 연결하는 직관적 학습 도구



2. 시장 환경 분석

디지털 전환 가속화에 따른 글로벌 에듀테크 및 바이오-인포매틱스 시장의 급격한 성장



* 글로벌 바이오-에듀테크 시장 연평균 성장률(CAGR) 약 21.4% 전망

3. 마케팅 계획 - 3C 분석



Customer

학부생, 고교 생명과학 동아리, 그리고 제약/바이오 기업의 신입 사원 직무 교육 수요층



Competitor

기존 정적 텍스트북 및 고가의 기업용 킵코 바이오웍스(Ginkgo Bioworks) 대비 우수한 가격 경쟁력



Company

서울대 생명과학부 연구실 기반의 검증된 데이터셋과 고도화된 AI 파이프라인 (SpliceAI+AlphaFold)

4. SWOT 분석

Strengths (강점)

- 자체 학습된 고정밀 SpliceAI 모델
- Mol* 기반의 직관적 3D 시각화
- SNU 연계 생물정보학 전문성

Weaknesses (약점)

- ColabFold 실행을 위한 고가의 GPU 비용
- 초기 단계의 제한적 유전자 라이브러리

Opportunities (기회)

- 디지털 트윈 교육에 대한 정부 지원 확대
- 개인 맞춤형 정밀 의료 시장의 확대
- 온라인 원격 실험 수요 증가

Threats (위협)

- 글로벌 빅테크의 교육 시장 진입
- 오픈소스 모델의 급격한 기술 변화

5. 마케팅 계획 - STP 전략

S

Segmentation
(시장 세분화)

학문적 깊이에 따른
K-12 / University / Professional

T

Targeting
(목표 시장)

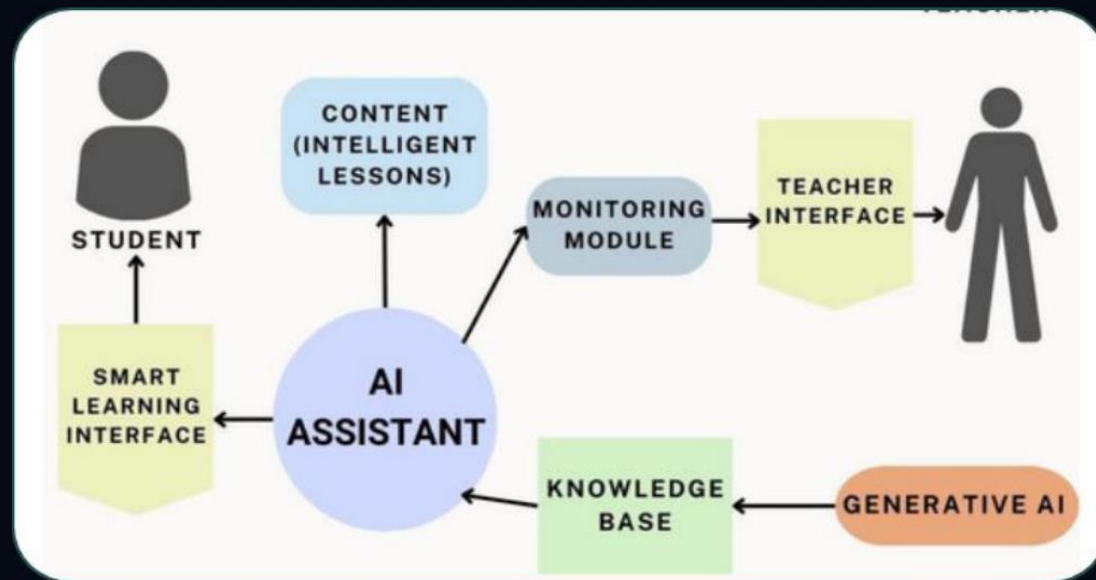
실습 장비가 부족한
지방 대학 및 특성화 고교

P

Positioning
(위상 정립)

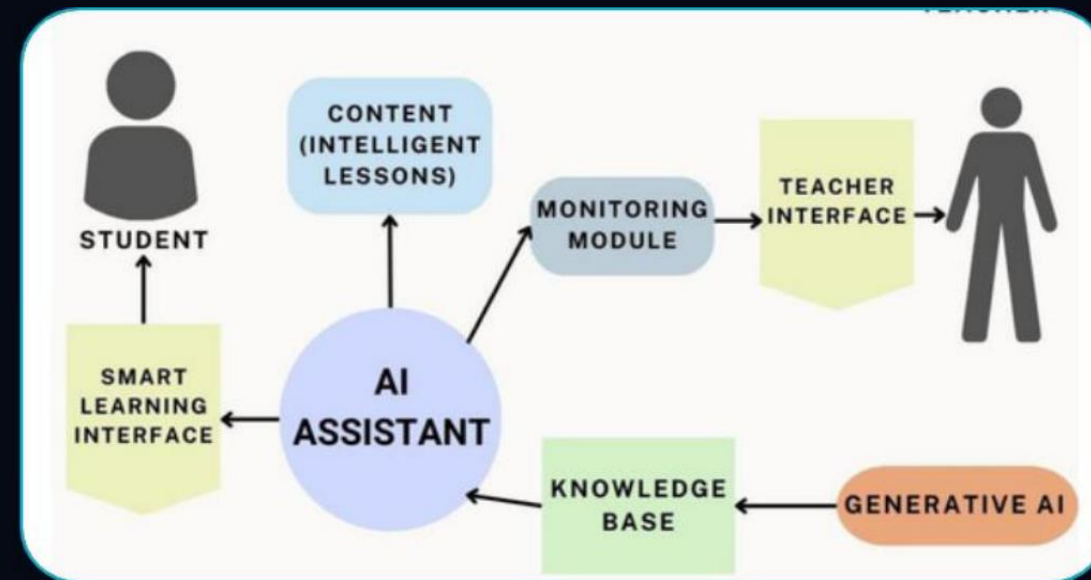
세상에서 가장 쉬운
'가상 유전체 실험실'

5. 마케팅 계획 - 4P MIX



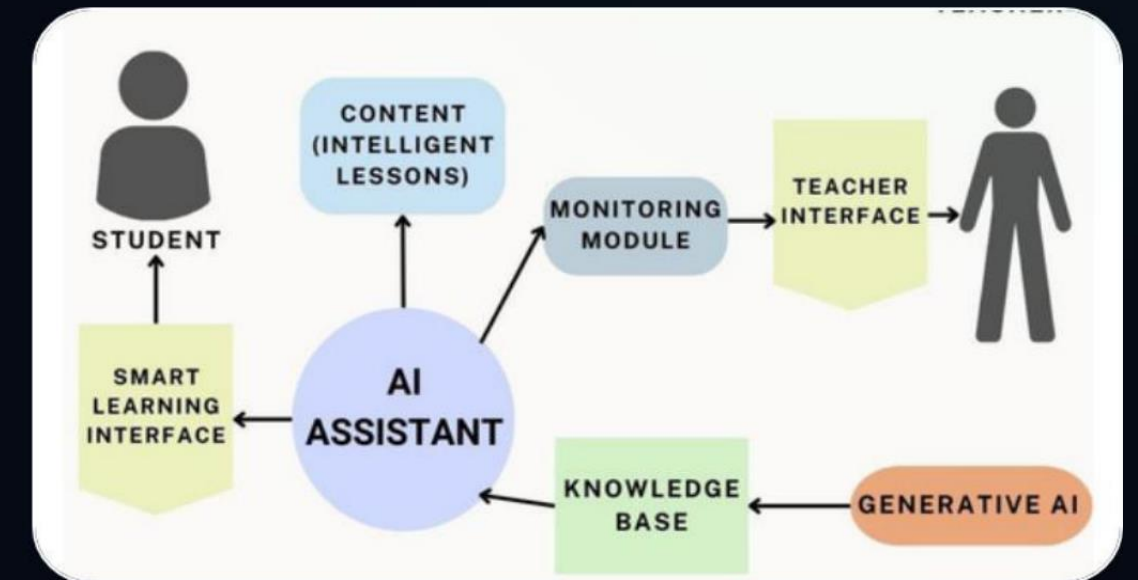
Product & Price

SaaS 기반 웹 서비스
B2B/B2G 구독 모델 도입



Place

글로벌 웹 플랫폼 배포
교육용 소프트웨어 마켓



Promotion

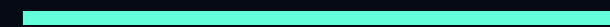
바이오 학술 대회 전시 참여
오픈소스 커뮤니티 홍보

6. 창업시기?..



Q&A

Splice Playground: Visualize Splicing Into Structure



경청해 주셔서 감사합니다.

AI01분반 4조 - 강민준, 남윤정, 이정균, 김현우, 최진범

IMAGE SOURCES



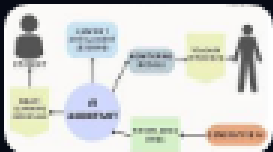
<https://cdn.the-scientist.com/assets/image/52569/lab-instrument-800x560-l.jpg>

Source: www.the-scientist.com



https://images.stockcake.com/public/7/e/a/7ea4f07c-ecd0-4260-bbf1-bc7afbc628c7_large/genetic-data-concept-stockcake.jpg

Source: stockcake.com



https://www.mdpi.com/applsci/applsci-15-05785/article_deploy/html/images/applsci-15-05785-g009-550.jpg

Source: www.mdpi.com

Reference

- Predicting Splicing from Primary Sequence with Deep Learning. Cell. 2019 Jan 24;176(3):535-548.e24.
<https://pubmed.ncbi.nlm.nih.gov/30661751/>.
- Jumper, J., Evans, R., Pritzel, A. et al. Highly accurate protein structure prediction with AlphaFold. Nature 596, 583–589 (2021).
<https://doi.org/10.1038/s41586-021-03819-2>.
- Some Lecture Notes from [Prof. Daehyun Baek](#) (School of Biological Sciences, Seoul National University)
- Some Lecture Notes from [Prof. Sun Kim](#) (School of Computer Science and Engineering, Seoul National University)
- <https://www.sciencedirect.com/science/article/pii/S1874939916302759>